

Replicating Manifold Steering: Geometry of LLM Representations and Behavior

Kirill Acharya · Stanford University · CS 221M Final Project

Code, checkpoints, and figures: <https://github.com/kirillacharya/manifold-steering-replication>

Background: do LLM representations and behavior share a geometry?

Large language models build up internal representations layer by layer: at layer L , each token maps to a hidden state $h^{(L)} \in \mathbb{R}^d$, and structured concept domains may organize into geometric patterns. Hidden states for *days of the week* or *months of the year* may trace a smooth, low-dimensional manifold $\mathcal{M}_h \subset \mathbb{R}^d$ — closed loops for cyclic domains, open curves for sequential ones. Wurgaft et al. [1] ask whether this geometry *predicts behavior*: output probability vectors, embedded via the Hellinger map $p \mapsto \sqrt{p}$ (which makes the Hellinger distance $d_H(p, q) = \frac{1}{\sqrt{2}} \|\sqrt{p} - \sqrt{q}\|$ Euclidean [2]), may trace a matching *behavior manifold* $\mathcal{M}_y$. If $\mathcal{M}_h \approx \mathcal{M}_y$ isometrically, steering *along* $\mathcal{M}_h$ should produce natural behavioral transitions, while linear steering cuts off-manifold and “teleports” probability mass. The two strategies interpolate between a source concept $\hat{h}_{c_s}$ and a target $\hat{h}_{c_e}$:

$$\pi_{\text{lin}}(\alpha) = (1 - \alpha) \hat{h}_{c_s} + \alpha \hat{h}_{c_e}, \quad \pi_{\text{mfd}}(\alpha) = s((1 - \alpha) u_{c_s} + \alpha u_{c_e}), \quad u_{c_i} = s^{-1}(\hat{h}_{c_i}),$$

where $s : \mathbb{R}^k \rightarrow \mathbb{R}^d$ maps intrinsic coordinates back to activation space. We replicate the paper’s central figure on a small open model, then extend it to a 2D concept structure (a day×hour torus).

The experiment, step by step

Setup. All experiments use instruction-tuned **Gemma-2-2B** ($d = 2304$), patching layer 24 of 26. Two cyclic concept domains: *weekdays* (7 classes) and *months* (12 classes), probed with arithmetic prompts — “Q: What day is two days after Monday? A:” (answer: *Wednesday*), 6 prompts per concept (`scripts/run_weekdays.py`, `run_months.py`). For each prompt we record the last-token hidden state $h^{(24)}$ and the output distribution over concept tokens, then

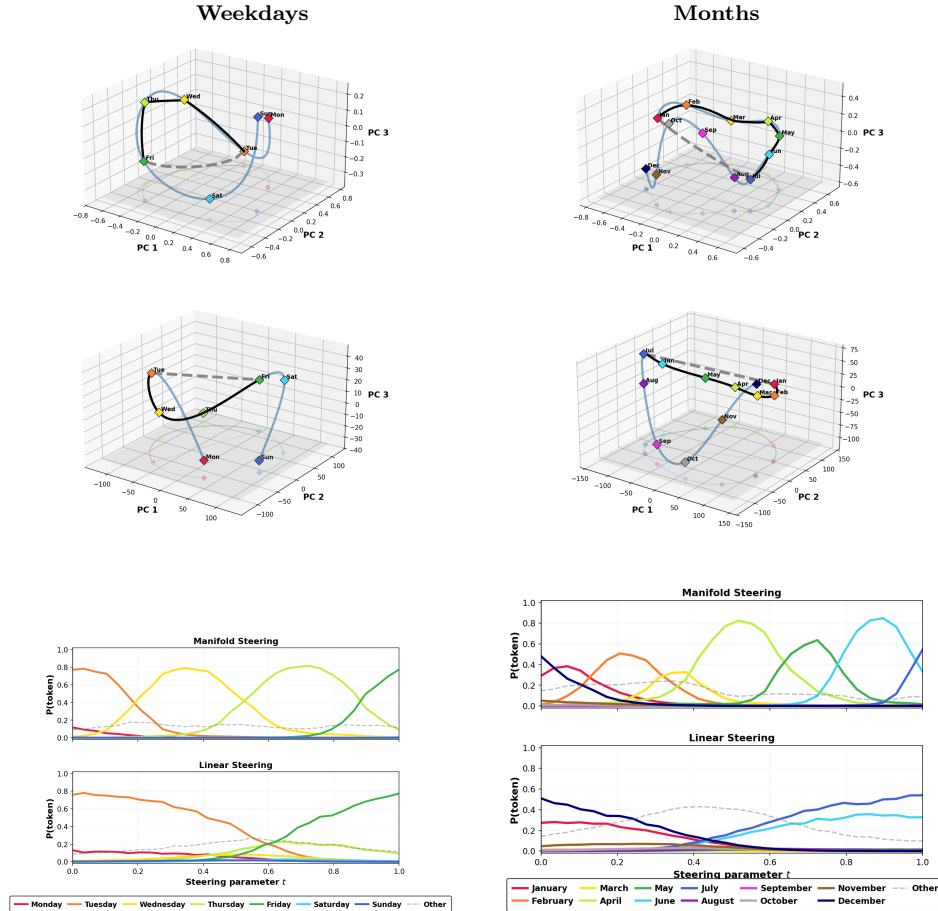


Figure 1: Gemma-2-2B layer 24, steering Tue→Fri (weekdays) and Jan→Jul (months). Rows 1–2: fitted behavior manifolds $\mathcal{M}_y$ (Hellinger–PCA) and activation manifolds $\mathcal{M}_h$ (activation–PCA) with the linear path (dashed) and manifold path (black). Row 3: concept-token probabilities along each path.

average per concept: $\hat{h}_c = \frac{1}{N_c} \sum_{i \in S_c} h_i^{(L)}$ and $\hat{p}_c = \frac{1}{N_c} \sum_{i \in S_c} p_i$.

Manifolds. A “manifold” here is simply a smooth curve through the concept centroids. The 7 weekday centroids $\hat{h}_{c_k} \in \mathbb{R}^{2304}$ span at most a 6-dimensional subspace, so PCA to $r=6$ ($r=11$ for 12 months) is lossless — it just re-expresses each centroid in coordinates $z_k \in \mathbb{R}^r$ adapted to the data. We then fit a cubic spline $\gamma: \mathbb{R} \rightarrow \mathbb{R}^r$ through the points *in concept order*, with the concept index as curve parameter: $\gamma(k) = z_k$ exactly at each knot, and between knots $\gamma(t) = a_k + b_k(t-k) + c_k(t-k)^2 + d_k(t-k)^3$, coefficients solved jointly so the curve is C^2 -smooth. This gives the activation manifold $\mathcal{M}_h$: a single scalar coordinate (“where between Monday and Sunday”) decodes to a full hidden state via $\text{PCA}^{-1}(\gamma(t))$. The behavior manifold $\mathcal{M}_y$ is built identically from the probability centroids, except each $\hat{p}_{c_k}$ is first mapped to $\sqrt{\hat{p}_{c_k}}$ (Hellinger space), where Euclidean tools — PCA, splines, distances — respect the geometry of the probability simplex.

Steering. For $\alpha \in [0, 1]$ between spline indices i_s, i_e : $\pi_{\text{mfd}}(\alpha) = \text{PCA}^{-1}(\gamma(i_s + \alpha \Delta i))$. At each of 30 steps, a forward hook replaces the layer-24 hidden state at the last token with $\pi(\alpha_t)$ and the output probability of every concept token is recorded (`scripts/run_steering_probs.py`). Everything is deterministic; all figures regenerate from committed checkpoints without a GPU (see the repo README).

Replication of the main figure

Figure 1 replicates the paper’s main result. In both domains the concept centroids trace closed, ordered loops in activation space and in behavior space — the cyclic structure of weeks and months is visibly encoded. The manifold path stays on the fitted curve and, in the probability trajectories (row 3), produces *smooth, ordered* transitions: steering from Tuesday to Friday passes probability mass through Wednesday and Thursday in sequence. Linear steering instead leaves the manifold and *teleports*: intermediate concepts never rise in order, and off-concept mass appears mid-path. One practical lesson: patch deep enough — at layer 12, keys/values from the original prompt override the patch over the 14 remaining blocks and both methods flatline; layer 24 (92% depth) is clean.

Extension: torus steering (day $\times$ hour)

Do LLMs encode *two-dimensional* concept structure? A weekday lives on one circle; a (day, hour) pair lives on a *product* of two circles — a torus. We collect layer-24 activations for all 168 prompts “It is HH:00 on {Day}” (7 days $\times$ 24 hours) and assign each prompt two angles, $\theta_{\text{day}} = 2\pi k/7$ and $\theta_{\text{hour}} = 2\pi h/24$. Each circle’s 2D subspace $V_{\text{day}}, V_{\text{hour}} \in \mathbb{R}^{2 \times d}$ is recovered by regressing centered activations onto the standardized labels $[\sin \theta, \cos \theta]$ — each label component yields one direction $d \propto \sum_i (h_i - \bar{h}) \ell_i$ — followed by QR-orthogonalization of the two directions (`scripts/run_hierarchical_steering.py`). The two recovered planes are nearly orthogonal to each other ($\max |\cos| \approx 0.18$), so day and hour occupy largely independent planes of the residual stream. Projecting each activation onto both planes gives two recovered angles, rendered on a torus — day on the major circle, hour on the minor (Fig. 2, left). The 7 day-arcs are cleanly ordered; the hour circle does not fully close, so hour is only partially circular here.

Steering in *product coordinates* respects this factorization: starting from the source activation h_{base} (Tue, 09:00), each step moves the day and hour coordinates *independently within their own planes*, $h(\alpha) = h_{\text{base}} + \Delta_{\text{day}}(\alpha) V_{\text{day}} + \Delta_{\text{hour}}(\alpha) V_{\text{hour}}$, where $\Delta_c(\alpha)$ interpolates the source $\rightarrow$ target offset within each plane — the 2D analogue of walking along the manifold. Each patched activation is decoded into a day $\times$ hour probability grid: day from token probabilities, hour by nearest centroid in the V_{hour} plane (two-digit hours are multi-token for Gemma, so subspace readout is more faithful). Steering (Tue, 09:00) $\rightarrow$ (Fri, 15:00) over six steps (Fig. 2, right), the bright cell walks diagonally — both coordinates move smoothly — while linear steering moves the day correctly but teleports the hour, jumping to 15:00 only at the final step: the same off-manifold failure mode, now visible per coordinate.

Takeaway. Representation geometry is not decoration: moving along it produces natural behavior, and ignoring it breaks transitions exactly as the manifold picture predicts. Code, checkpoints, SLURM scripts, and a no-GPU figure regeneration path: <https://github.com/kirillacharya/manifold-steering-replication>.

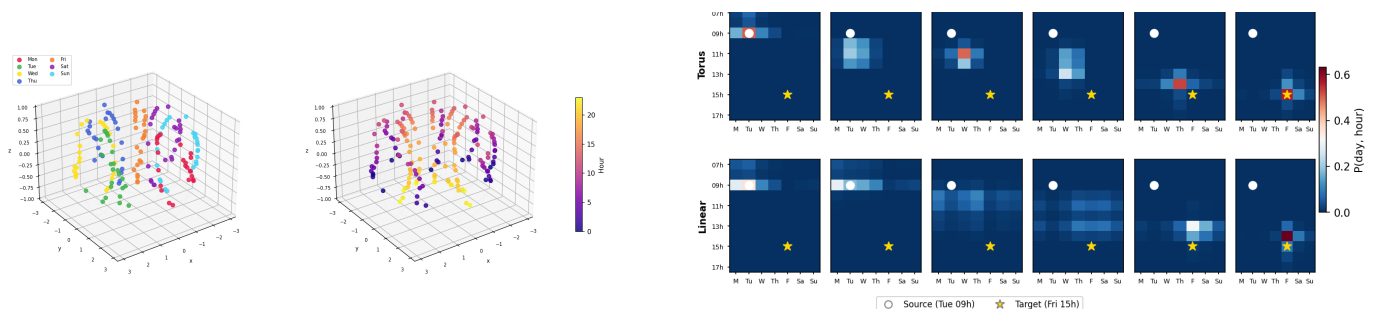


Figure 2: Left: layer-24 activations on the day $\times$ hour torus, colored by day (7 ordered arcs) and by hour. **Right:** day $\times$ hour probability grids across 6 steering steps, (Tue, 09:00) $\rightarrow$ (Fri, 15:00); top row torus (product-coordinate) steering, bottom row linear steering.

References

- [1] D. Wurgaft, C. Rager, M. Kowal, V. Shyam, S. Feucht, U. Bhalla, T. Haklay, E. Bigelow, R. Sarfati, T. McGrath, O. Lewis, J. Merullo, N. D. Goodman, T. Fel, A. Geiger, and E. S. Lubana. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv preprint arXiv:2605.05115*, 2026.
- [2] S. Amari and H. Nagaoka. *Methods of Information Geometry*. American Mathematical Society, 2000.